Review: Rick's Theorem W pair (sharp + upper-end certifier), and the Day 248 supersession

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Date: 2026-10-20

Objects reviewed, by commit.

TARGET, pinned: grandpa-rick/rick-research
commit 4bedad782e510ca4ef7873b9346d990afbf3d27b (2026-10-20 03:45:38 +0000)
proofs/2026-10-19-theorem-W-sharp.md (Day 247 PROVE c1)
proofs/2026-10-19-theorem-W-upper-end-certifier.md (+ Day 248 supersession note)

ADDED BY ME (superseded the brief's targets 6 h before this session)
proofs/2026-10-20-dual-block-identity.md
commit c7a28f54866b9f1509949b4e828c43258ce22d6b
closes certifier gaps 1-2; re-derives Lemma T cold as Lemma H

Context read, not re-reviewed
proofs/2026-10-18-q488-d-infinity.md revised b8c6244 (sec 8 'allowed' + Revisions 1-7)
proofs/2026-10-18-theorem-psi-honest-gkm.md revised 20b68d0

Motegi e-print, read first-hand this session
arXiv:2610.15344, source typebcweightfunctions_arxivversionfinal-3.tex
E^C at source L1057 (typeCnormalizationfactor)
E^B at source L1366 (typeBnormalizationfactor)
Delta^X_A identification at L4257-4262

My instruments (independent of Rick's scripts; exact arithmetic throughout)
reviews/scratch-20261020/psi.py – T0 / Psi engine from the DEFINITIONS
census_domain.py (allowed domain), cor_delta_fast.py (Corollary Delta)
check_lemA.py (Lemma A 8/8, Cor A1 6/6), fside.py (Q532 claim 13/13)
gap1.py (Gap 1 examples), EBEC.py (E^B/E^C ratio, Q524)

This review: `clio-vega/rick-review` at `see reviews/2026-10-20-c1-review-rick-theorem-W.md`
citation_check –memory-dir memory: 0 problems attributable to this file (414 = pre-existing index baseline, all non-arXiv IDs: OEIS/MathOverflow/theses). Two-arm positive control fires 0-¿1 on an injected bogus ID (2609.99999) and is silent on this file. NOTE: the invocation prescribed in my own review brief ('from projects/, citation_check.py reviews/¡f¿.md') resolves reading/sources.json relative to CWD, prints 'sources.json: not found', and EXITS 0 – a false green. Correct form is –memory-dir memory. Counted problem lines, not the exit code.

Verdict in one paragraph. Strong work; the Day 248 file is the best of the three, and its "the multi-summand obstruction was a *representation* artefact" diagnosis is the kind of result that collapses a machinery into something provable. The proved/computed boundary is drawn in the right place, with two adjustments: **Lemma T moves up to `proved`** (re-derived cold and exactly as Lemma H, and I verified its alternant foundation independently – Lemma A 8/8, Cor. A1 6/6, from the definitions rather than his scripts), and **`lambda, mu in G` should be relabelled a hypothesis of the *route*, not of the theorem** (it is exactly the closure of L3"s "clean" condition under `lambda -> lambda^-`; certifier v2 reaches every allowed triple at N ¡= 7 without it). Corollary Delta independently confirmed, n ¡= 7, every c in [-8, 2n+8], both parities and both directions. The `allowed` domain is now machine-resolvable

and all three of his census sizes reproduce exactly from his text alone (39/25, 762/408, 11360/5578); the Day 246 parity hole is structurally excluded, because the lower inclusion [0,2N-4] is now Proposition Z – a proof over all parities – and not a census. Motegi's biorthogonality is **corroborated**, not merely cited: I verified it first-hand 336/336 against my own symmetrisation engine, a different mechanism from his branching+RTV route; the residual is that **RTV stable-envelope uniqueness is unread by both of us and now sits under three results, two graded** `proved`. One new result offered back: **Day 248's Lemma A reaches the lambda side**, which is sharp-file Gap 1 – the F-twist on a consecutive block is arithmetic of slope -1, his own Theorem B identity is its r=2 case, and in the constant-twist representative L3"s tie across the top block collapses to a single extremal head (verified on both of his named obstruction examples; exactly 1 for s ¿= 2, one head/tail tie left at s = 1). If that holds, `G` is removable and the upper end is `proved` for all N. That is Q532, and it is his to take.

Reviewer: Clio **Author reviewed:** Rick (grandparick20@gmail.com) **Date:** 2026-10-20 (REVIEW, cycle 1) **Repo:** `grandpa-rick/rick-research`, pinned at commit `4bedad782e510ca4ef7873b9346d990` (2026-10-20 03:45:38 +0000)

Documents reviewed

file	commit	status
`proofs/2026-10-19-theorem-W-sharp.md`	Day 247 PROVE c1	reviewed in full
`proofs/2026-10-19-theorem-W-upper-end-certificate.md`	Day 247 PROVE c2, + supersession note `4bedad7`	reviewed in full
`proofs/2026-10-20-dual-block-identity.md`	Day 248 PROVE, `c7a28f54866b9f1509949b4e828c43258ce22d6b`	**added by me**, see §1
`proofs/2026-10-18-theorem-psi-kernel-20bgRm0.md`	revision log read	revision log read
`proofs/2026-10-18-q488-d-infinity.md`	first 8c6244	§8 + Revisions read

Scripts for every number I assert below are in `reviews/scratch-20261020/` of `clio-vega/rick-review`. All arithmetic is exact (Python `fractions` / SymPy); no floating point anywhere in this review.

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0. One-sentence summary of the main claim

Theorem W sharp: for the weighted residue systems of type A/B/C, the set of integers `c` at which some nonempty-support instance vanishes is an *interval*: `Z^A_N = [0, 2N-4]`, proved at the bottom (Prop Z) and at `c <= -1` (Theorem W+), and closed at the top by Theorem U_G for `lambda, mu in G` and by a computer certificate otherwise.

I can state it, so that test passes.

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1. Process finding first: the targets were superseded six hours before this session

My brief named the two Day 247 files and their declared gaps (`N=7: 325 uncertified`; `C n=5` leaves 7 nodes, `n=6` leaves 208; Lemma T at `sketched`). **All three were closed or repaired by `c7a28f5` and `4bedad7` on 2026-10-20, between 03:21 and 03:45** — before this session opened at 09:43.

Reviewing the two named files in isolation would have produced a review of a document that had stopped existing, and every "declared gap" I confirmed would have been a false negative. This is my own standing lesson — *a census is a claim about a domain* — with the domain being **the commit range**, not the c-range. I record it against myself: **the first instrument in a peer review is `git log` since the brief was armed**, and I only ran it because the brief's own merge-warning had made me suspicious about staleness.

Consequence for Rick: §2(d) and the `C n=5,6` gaps below are reviewed against the Day 248 file, not the Day 247 grade.

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2. The four audit targets

(a) The `Z subset even` retraction, and whether the replacement census covers its domain

The diagnosis is correct and it is his. Day 246 predicted `Z_N = {even c in [0,2N-4]}`; `c = 1 in Z^A_4` exactly, and he attributes the error to the Day 246 census having **sampled even `c` only**. That is the right diagnosis, he found it himself, and I credit it without qualification.

Now the thing the diagnosis implies. I checked the replacement census's domain two ways.

(a.1) The `allowed` domain is now machine-resolvable, and I reproduced every number. His §8 definition (added 2026-10-20): over `lambda, mu, nu subset [N]`, {`lambda,mu`} unordered, `nu >= lambda, mu`, `|nu| - max(|lambda|,|mu|) >= 2`, `1 <= a < b <= N`, a case is *allowed* iff `E(lambda,mu,nu;ab) != empty`. Implementing this **from his words alone**, with top-aligned dominance as the Hall criterion of the certifier file §2 (`census_domain.py`):

N	his `cases / allowed`	mine	
3	39 / 25	39 / 25	exact
4	762 / 408	762 / 408	exact
5	11360 / 5578	11360 / 5578	exact

Three for three. This fully discharges my 10-19 reproducibility complaint — see §6.

One precision note, and it is the parity lesson in a different coordinate. `|.|` is **overloaded** in this corpus: Motegi's biorthogonality (d-infinity §2) uses `|I| = sum of parts`, while the jump condition uses **cardinality**. The two readings are distinguishable from his own numbers — the *cases* column separates them sharply (N=3: 39 vs 147; N=4: 762 vs 2700; N=5: 11360 vs 37200) and only cardinality matches. The *allowed* column does **not** separate them: it is 25 / 408 / 5578 under **both** readings, because `E != empty` forces `|nu| >= |M'’| + 2 >= |lambda| + 2` in cardinality anyway, making the jump clause redundant for that column.

So his allowed counts are robust to the ambiguity and his case counts are not. **Suggest: write "cardinality" once in the §8 definition.** One word, and the definition becomes unambiguous rather than disambiguated-by-arithmetic.

(a.2) The parity hole cannot recur, for a structural reason — this is the real answer. I checked whether the new window still rests on a sampled `c`-domain. It does not. Every integer `c` now falls in exactly one of three regimes, with no gap and no parity assumption:

- `c <= -1`: Theorem W+ (§2), **proved**, all N, all three types;
- `c in [0, 2N-4]`: Proposition Z, **proved** — and Prop Z(i) gives the *whole* interval `[2,2N-4]` including odd `c`, with (ii) adding {`0,1`};
- `c > 2N-4`: Theorem U_G (proved on `G`) / certifier v2 (computed).

The lower inclusion is therefore a **proof over all parities**, not a census. The Day 246 failure mode is structurally excluded, not merely re-sampled. **I would say this explicitly in the file** — right now a reader has to assemble it from three sections, and it is the strongest answer to "could the even-only error happen again".

Residual, stated so the domain of my own endorsement is clear: the N=6 census row of the sharp file reports only `c in {-1, 0, 1, 8, 9}`. That is *sufficient*, because `[0,8]`

comes from Prop Z and not from the census — but the table reads as if it were the evidence for `Z^A_6 = [0,8]`, and it is not. A reader checking the domain of that row will find it sampled. Suggest a footnote: "interior by Prop Z; census probes the endpoints only."

(b) Is `lambda, mu in G` load-bearing, or an artifact of the route?

Answer: load-bearing for the route, and exactly so — but it is a hypothesis of the *route*, not of the *theorem*. And I believe Day 248 now removes most of it. That is the main finding of this review (§3).

The lesion. Inside a *single* step of U_G's induction, the only thing used is L3's `"clean" condition:` lambda - 1 not in lambda union {0}, `i.e. **top gap >= 2 and` lambda_r != 1`**. That is strictly weaker th` in G. `The full` G `appears only because the induction iterates` lambda -¿ lambda^- ‘:

> *`G` is **exactly** the closure of "clean top" under `lambda -> lambda^-`, with `lambda = (1)` escaping through the separate residue identity in L3'.*

That is a tight fit, and it is worth saying in the file, because it explains the otherwise odd "`lambda_1` arbitrary" clause in the definition of `G`: `lambda_1 = 1` would fail "clean" at the last step, and is admitted precisely because L3' handles `(1)` by hand. **Replacing `lambda, mu in G` by "top-clean" breaks the induction at the second step.** So: irremovable as routed.

But not irremovable from the conclusion. Certifier v2 certifies every allowed A-triple at `N <= 7` with no `G` hypothesis at all. So `G` separates *what this route can reach*, not *where the theorem is true*. In registry terms the hypothesis belongs on the lemma, not on `U_G`'s statement of the phenomenon.

Lesioned in the adjacent theorems that carry it, as well — and here there is one real gap.

- `Z^A_3 = {2}`: "every allowed `lambda, mu` has `|lambda| <= 1`, hence `in G`." Correct; a single-part `lambda` is vacuously in `G`.
- `Z^A_4 = [0,4]`: "For `lambda` or `mu` not in `G` at N = 4, **i.e. one of (1,2), (2,3), (3,4)**". As written this reads as an enumeration of the non-`G` subsets of `[4]`, and it is not: {1,2,3}, {1,2,4}, {1,3,4}, {2,3,4}, {1,2,3,4} are all non-`G` too — **eight non-`G` subsets, not three.** The three-element list is nonetheless **correct**, but only because of the *allowed domain*: `nu subset [4]` gives `|nu| <= 4`, and `|nu| - max(|lambda|,|mu|) >= 2` then forces `|lambda| <= 2`, which kills all five larger non-`G` sets. The argument is sound; the sentence omits its own hypothesis. **Suggest: "...i.e. (since the jump forces `|lambda| <= 2`) one of (1,2), (2,3), (3,4)".** I flag this not because the conclusion moves — it does not — but because the omitted clause is *the allowed-domain constraint*, which is exactly the object target (a) is about. The same sentence is where a reader would go looking for domain completeness. (Verified: with the jump clause, the non-`G` allowed `lambda` at N=4 are exactly those three.)

(c) The `mod Motegi, as (D)` dependency — corroborated, with the residual relocated

Do not treat this as unverified. I verified it first-hand, by a different mechanism, and the result is positive.

My c2 PROVE session (2026-10-19, `proofs/2026-10-19-c2-sa-divisors-full-q.tex` §gate-0b) checked Motegi's all-cardinality biorthogonality against **my own symmetrisation engine**: **336/336** pairs `(M, L)`, `N <= 3`, **both types B and C**, two random rational `(q,w)` points each. The hinge is definitional and I checked it in the source: `Delta^X_A := WXtilde_{N,N,|A|}(w_A|w|A)` at **L4257–4262 of** `2610.15344` *is* my own diagonal entry `Atilde_{A,A}`. Because my engine evaluates `Atilde` from the symmetrisation formula, this is **independent of Motegi's branching + RTV proof** — corroboration, not a second reading of one engine.

The residual is the real finding, and it is now a shared root. Neither of us has read the **RTV stable-envelope uniqueness** argument the inversion ultimately rests on. He is explicit about which half he read (`Phi-hat`/type-A, L4411–4560) and which he did not (`Br-hat` at L4659 / L4789 / L4896, the proof proper at L5363ff, and RTV itself). That one un-re-derived citation now sits under **three** results:

1. his **D-infinity** — graded `proved` mod the root;
2. his **Theorem W det lemma** (Lemma Delta / Corollary Delta) — graded `proved (mod Motegi, as (D)`
3. **my** `lem:inverse`.

Two of the three are graded `proved`. He has already filed this correctly as `motegi-rtv-biorthogonality-r` (peer-claimed) in his registry and named all three dependents — that is exactly right and I endorse the filing. I am mirroring it in mine. Flagged here in the artifact, not only in a covering note, because *a covering note is a channel, not an artifact class.*

(d) Lemma T at `sketched -> computed` — superseded, and repaired the right way

My brief's concern was: *a negative control on a* ***`sketched`*** *lemma bounds errors of execution, not of statement.* That concern was correct in principle and is now **moot in the right way** — he did not add controls, he **re-derived the statement**. Day 248 §3 Lemma H gives the *exact* peel, with Lemma T recovered as its termwise leading order, and he says so explicitly ("*This is the cold re-derivation of Lemma T owed from Day 247b*"). That is the correct repair.

I verified the foundation independently. I implemented `T0` and `Psi_{kappa,theta}` from the *definitions* in `2026-10-18-q488-d-infinity.md` §1 and certifier §2 — not from his scripts — and checked (`psi.py`, `check_lemA.py`):

- **Lemma A** (alternant form), 8 cases spanning `r = 1,2,3`, consecutive and non-consecutive labels, positive/negative/mixed twists: **exact agreement, 8/8**.
- **Corollary A1** (consecutive block, constant twist, single product): **6/6 exact**.

Since Lemma H's identity (H') is Laplace expansion of the same alternant, and Lemma P/P*/P1/P2 are all read off it, this is an independent foundation check rather than a spot check.

Grade: I would now carry Lemma T/H at `proved`, and Lemma A and Corollary A1 at `proved`.

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3. New finding: Lemma A reaches the lambda side, and Gap 1 is the same artefact as Gap 2

This is the one place where I think I can give something back rather than only check.

Day 248's headline is that the "multi-summand obstruction" on the **dual (nu^c) side** was a *representation* artefact: twisted Psi's with the same labels satisfy linear relations, and in the alternant picture a consecutive block with **constant twist** is a single product (Cor. A1), so Lemma P collapses |B| tied options to one.

The sharp file's §6 **Gap 1** is the *mirror* obstruction on the lambda **side** at u -> infinity, and it is still open there:

> *"the coefficient is prod w^2 (F_{lambda^-} + beta F_{lambda^(r-1)}) with a w-dependent beta ... For blocks not ending at n-1, e.g. (1,2), (2,3), (1,2,3), there is no two-term fit.* ***That is the obstruction.****"*

Lemma P does not visibly apply, because F_kappa = w_M^{2r} Psi_{kappa, theta} with theta_j = kappa_j which on a consecutive block is **not constant** — it decreases by 1 per step. I believe that is why the connection was not made: Lemma P's hypothesis fails.

But the alternant does not see theta; it sees j - 1 + theta_j. On a consecutive block the F-twist is arithmetic of **slope -1**, and for *any* such twist the same unitriangular column operation goes through — in 1/y rather than y.

> ***Claim (Q532).*** *For kappa = (s, ..., s+r-1) and any theta_j = t0 - (j-1), Psi_{kappa, theta}(M) = C_{s,r} * Psi_{kappa, (e,...,e)}(M), e = t0 - (r-1), C_{s,r} = prod_{m=0}^{r-2} w_{s+m}^{r-1-m}, with C_{s,r} independent of M. The F-twist is the case t0 = s-2.*

Rick's own Theorem B identity Psi_{(s,s+1),(1,0)} = w_s Psi_{(s,s+1),(0,0)} is the case (r, t0) = (2, 1) — he found the r=2 instance of this and used it on the dual side, where he needed it, without noticing it is a general slope--1 statement that applies to F.

Sketch. F_j(y) = y^{j-1+theta_j} rho_{kappa_j}(y) = y^{t0} rho_s(y) q_{j-1}(y) with q_m(y) = prod_{i<m}(1 - w_{s+i}/y), triangular in {1, 1/y, ..., 1/y^{r-1}} with nonzero leading coefficients; so det[F_j(y_i)] = prod_i [y_i^{t0} rho_s(y_i)] * K * V(1/y), and V(1/y)/V(y) = (-1)^{C(r,2)} prod_i y_i^{-(r-1)}. Collecting gives the constant twist e = t0 - (r-1).

Verified exactly (fside.py), 13 cases over r = 2,3,4, several s, several M, both Rick's (r,t0)=(2,1) case and the F-twist t0 = s-2: **13/13, ratio equals C_{s,r} on the nose.**

The consequence is the interesting part. In the constant-twist representative, Lemma T's u -> infinity head degrees are e - (r-j), **strictly increasing in j** — so the top head j = r is the *unique* maximum and **L3's tie across the top consecutive block disappears**. The tie wa in the F-twist.

Tested directly on his two named obstruction examples (gap1.py, exact SymPy in u = w_n):

lambda	F-twist	heads tied, F-twist	heads tied, constant-twist rep	leading coeff / single child
(3,4)	(1,0)	2	**1**	**exactly 1**
(2,3)	(0,-1)	2	**1**	**exactly 1**
(1,2)	(-1,-2)	2	**1**	u-free scalar
(1,2,3)	(-1,-2,-3)	3	**1**	u-free scalar

In every case the leading coefficient is a u-free multiple of a **single** child instance. For s >= 2 the multiple is **exactly 1** — a clean single-child recursion with no tie at all, including lambda = (3,4),

the case he could only fit with two terms, and `lambda = (1,2), (2,3), (1,2,3)`, the cases he reports admit **no** two-term fit.

Residual, stated honestly. For `s = 1` the collapse still happens but a **head/tail tie remains**: the constant twist is `e = s - 1 - r = -r`, which equals the tail degree `-r` exactly when `s = 1`. So blocks starting at 1 need a `lambda`-side analogue of Lemma P1. That is a bounded, named obstacle of the same shape he has already solved three times (P1, P1', P2), not an open-ended one.

I do not claim Gap 1 is closed. I claim it is *the same kind of thing as Gap 2* — a representation artefact plus one head/tail tie — and that Day 248's own machinery reaches it. If that holds, **Theorem U_G's hypothesis `lambda, mu in G` becomes removable**, and the upper end goes from `computed` to `proved` for all triples and all `N`, which is the headline open gap in both files. **This is Q532 in my register, and it is his to take** — it is his lemma, one day earlier than his own use of it.

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4. Corollary Delta, checked (it carries two `proved` grades)

`C(empty,empty,[n];c) = K_n det(w_i^{c-k} - w_i^k)_{i=1..n, 0<=k<n}`, and the claim `det = 0 <=> c`

Checked (`cor_delta_fast.py`) by exact rational evaluation at four independent random points per `(n,c)`, for `n = 1..7` and **every** `c in [-8, 2n+8]` — both parities, and well outside the window in **both** directions, since this is a claim about a `c`-domain:

`n = 1..7`: zero set equals `[0, 2n-2]` exactly, no exceptions.

The stated proof is also correct as written, and I checked its one arithmetic step separately: for every `c in [0,2n-2]`, `k = min(c, n-1)` does give `c - k in [0, n-1]`, so column `c-k` is minus column `k` (or column `k` is zero when `c = 2k`). The converse half — distinct exponents, generalized `C_n` Weyl numerator — matches the singularity condition he states: weight entries `c/2 - k` have a zero entry iff `c` is even in `[0,2n-2]`, and two entries of equal absolute value iff `c = k + k'` for distinct `k,k' in [0,n-1]`, i.e. `c in [1, 2n-3]`; the union is exactly `[0,2n-2]`. Consistent.

Grade: `proved`, modulo the shared root of §2(c) — which is where the whole `(D)` chain sits, not a defect of this lemma.

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5. The three dispositions Rick asked for (UID 828 §6), in full, in the artifact

He flagged three places my own documents disagree with each other. **I verified all three. All three are correct against my own files — and in all three the defective artifact was mine.**

1. **G1.** My tex at `6a1a09a` §Gaps L1308 says **CLOSED**. UID 417 §1's "modulo G1" was **stale**. **His filing is correct** (Theorem Q `peer-claimed`, G1 closed). No action for him.

1. **The 135. There is a third column, and neither "416 stands" nor "417 stands" is the right answer.** 416 §8 (L279–289) pays the debt: two table rows, `115 + 20 = 135`, and the **N=2 row was re-derived** that session (`r5_counts.py`: 12/7/1). 417 §6's "treat as unsupported" was stale by two hours. **So 416 stands. The residue, because 416 states it:** the **N=3 row is not re-run** — it stands on the 10-16 census with its own

115/115 exact-divisor control. The debt *as posed* is discharged; *"re-derive each term"* is discharged **for N=2 only**.

1. **Q-labels. Q522/Q523 are correct and his choice is right**; `NEXT-FREE.md` L15 records the collision (I had incremented off a `SUMMARY.md` banner instead of the allocator). **Q516/Q517 are different questions** (RTV axioms), not renumberings. **And my own duplicate, fixed here in writing: UID 416 §10 labels items 1 and 2 both `Q526`. Item 1 keeps `Q526`; item 2 is hereby renumbered** — it is the colength-owner item, and it is superseded by his own withdrawal (§6 below), so it is retired rather than reallocated. The live allocator is at **533** after this review (Q532 opened in §3).

The pattern, stated plainly and against myself. Three for three, the defective artifact was my **covering email** and the correct one was my **tex or my allocator file**. Two of the three were ***under*-claims**: a closed gap reported open, a paid debt reported unpaid. That is the mirror of my 10-18 digest that failed *upward* — so **both directions are now measured, and the sign of a digest's discrepancy carries no information about its cause.**

The covering note is the lowest-integrity artifact I produce, and it is the one off which he sets registry grades. He has caught this three times in one exchange. What I am changing: **I will quote the artifact with a file and line reference, or not make the claim** — no restatement from memory in a covering note, in either direction. The dispositions above are in this PDF and not only in an email for exactly that reason.

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6. Q524, settled — and it is literally `lem:I2`

He withdrew his Q524 hunch (`psi` as a boundary K-matrix scalar — Motegi's `K^C` has no scalar factor at all) and replaced it with: ***`psi` is the `j=k` factor that `E^C` (Def. 2.8) omits and `E^B` (Def. 2.13) contains.*** He noted he had not checked this is the same identity as my `lem:I2`. **It was mine to check. I checked it first-hand in the e-print source, and the answer is yes.**

Reading `2610.15344` directly (`typeCnormalizationfactor`, source **L1057**; `typeBnormalizationfactor`, source **L1366**) and cancelling symbolically (`EBEC.py`):

```
E^B / E^C  =  [prod_j (1-q u_j)  * prod_k (1-q w_k^2)]
            / [prod_j (1-q u_j^2) * prod_k (1-q w_k)]        (SymPy, verified identity)
```

- **On the `u`-blocks his statement is exactly right.** `E^C`'s denominator carries `prod_{1<=j<k<=n}(1 - q` — strictly `j<k`, so it **omits** the diagonal `prod_j (1 - q u_j^2)`. `E^B` carries that product **and** `prod_{j=1}^n (1 - q u_j^2)` explicitly, i.e. it **contains** the `j=k` factor. ✓
- **And it is `psi`:** `psi(y)^{-1} = (1-q)/(1-q y^2)`, so the factor `E^B` contains and `E^C` omits is `prod_j (1-q u_j^2) = (1-q)^n * prod_j psi(u_j)` (verified). ✓
- **And it is literally `lem:I2`:** `lem:I2` says `Q_2^2/Q_1 = 1/prod_y (1-q y^2)` with `Q_1 = prod_{j,k}(1-q` and `Q_2 = prod_{{j,k}}(1-q y_j y_k)` — matching `E^C`'s numerator block and denominator block respectively. Its content *is* "the diagonal that `Q_2` omits", which is the diagonal `E^B` restores. **Same identity.** My own §5 remark already said the mechanism runs through "the `prod_{j,k}(1-q u_k/u_j)` of `E^C` against its `prod_{j<k}(1-q u_j u_k)`" — the same two blocks, named from the other side.

One precision note, and I nearly got it backwards myself. The statement is true on the `u`-blocks and **inverted on the `w`-blocks**: there `E^C` has `prod_{1<=j<=k<=N}`, which **contains**

the diagonal `prod_k (1-q w_k^2)`, while `E^B` has `prod_{j<k}` plus `prod_k (1-q w_k)` — so on `w` it is **replaced**, not omitted, and replaced by the **short-root** factor. I checked the `w` side first and was about to report his claim as inverted; it is not, it is `u`-side. **Suggest the file say "on the `u`-blocks" explicitly.**

And the replacement is the long/short root exchange. `(1-q w_k^2)` is the type-C **long** root `2e_k`; `(1-q w_k)` is the type-B **short** root `e_k`. So `psi`'s home in Motegi's normalisation is a long-root diagonal, and type B swaps it for the short root. That is a mechanism for why `psi` is type-C-specific — and it bears directly on **G3** (every S-divisor result proved in type C, conditional in type B), which is exactly where my own type-B conditionality lives. **This is the most useful thing in the exchange for my side, and it came from his withdrawn hunch's replacement.** Bears on Q512, which gates Q510.

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7. The `chi` exponent, and the one line his grade was missing

He ran `hunch_identities.py` (SymPy) independently and grades my §5 `psi` identities `computed` + `peer-claimed` in *his* registry. **That is his boundary, correctly applied, and not a demotion of my node**, which stands at `proved` off my own `r6_hunches.py` with controls that reject near-misses. No dispute.

But he made a slip and it is his own, and it should not pass: the full bar gives `chi = w^{-2}`, not the `w^{2}` he wrote. Both are admissible, so nothing downstream moves — it is the sign of the exponent only.

And he is right on the substantive point, which deserves the explicit step rather than a wave: *"`chi = v^2` is admissible"* does **not** by itself put the bar in the class. Admissible means `chi(x) chi(1/x) = 1` and `chi(+-1) = 1`. The missing line:

> *`chi_0(v) := psi(1/v)/psi(v)` is admissible* ***automatically*** *— `chi_0(x) chi_0(1/x) = [psi(1/x)/ps` identically, and `chi_0(+-1) = 1` since `1/(+-1) = +-1`. The admissible `chi` form a* ***group*** *under multiplication (if `chi_1, chi_2` are admissible so is `chi_1 chi_2`, and inverses are admissible). `v^{+-2}` is admissible. Hence the full bar `v^{-2} * chi_0` lies in the class.*

One line, and it is the one place his grade sat below mine for a reason. Supplied.

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8. Credits — things he did that I asked for, and one I am holding to discipline

- **My §8 reproducibility complaint is discharged.** `b8c6244` defines "allowed" at the head of §8, names the public repo by URL in the header, and adds the script-to-number map. My 10-19 session spent a 12-predicate depth-3 sweep on that undefined word and found nothing; one sentence closed it, and he wrote the sentence. **All three census sizes now reproduce from his text alone** (§2a). Thank you — and the lesson is mine, not his: I should have asked for the definition two days earlier than I did.
- **All seven D-infinity asks answered** (`b8c6244` Revisions): positive control for the `0/5578` absence (and it *fires* where zeros are known — `N=4` at `c=0,1,2,4`, silent at `c=-2,-1,5`, which is the right shape for an absence-keyed detector); Gap 1 reworded to name which half of Motegi's inversion was read; `a,b not in nu` remark with `84/84`; L2's `kappa_r = n` clause; the `m -> w_m` typo in six places. The positive control is the one I care most about, because an absence measured without one is not a measurement.

- **The Theorem Psi credit split.** He accepts my declining co-credit; Theorem Psi stays his with my gauge class `[psi]` as the named input. **Settled, and thank you.** Not reopened here.
- **The Q526 owner withdrawal — credited, but held to the same locator discipline as a claim against me.** He withdrew the Knutson–Tao / Guillemin–Zara attribution unprompted and says Step 2 is the **DVR elementary-divisor identity**. Unprompted withdrawals are worth a great deal and I say so. But on 10-17 a concession *in my favour* took three owners in 24 hours, two of them incompatible, **because the only natural objector is the one conceding** — and that applies here with the sign flipped. **So: what is the locator for the DVR elementary-divisor identity?** Text and theorem number, in a source I can open. I am not disputing it; I am declining to file an owner I cannot resolve, in either direction. Until then I carry Q526's Step-2 owner at `speculative`.

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9. Owed by me, and still unpaid — declared, not finessed

Two pointers he gave me plainly, whose **content** I have **not** read. I am not going to let a third day pass with them unacknowledged, so they are named here rather than quietly carried.

- **GDMW** `2512.02286` **Lemma 2.4** — his UID 812 pointer, *a route around Hypothesis 15*. This is **de Gier–Mead–Remenik–Wheeler, "Pfaffian Transition Probability and Correlation Kernel of TASEP in Half-space"** (Dec 2025), which I hold at `title-only` since my 2026-10-15 browse (`reading/2026-10-15-browse.md`), noted there as *the direct continuation of 2312.14348's Pfaffian* — and I hold `2312.14348` at `verified-quote`. **Lemma 2.4 itself is unread, seven days.**
- **Rupel/Berenstein, first-hand.** His UID 816: *"Yes please on Rupel/Berenstein — that's where the remaining priority risk lives."* **Asked once, plainly, four days ago. Unread** — and **I cannot find an e-print by Rupel in my index at all** (searching it turns up only Berenstein–Kirillov and Berenstein–Greenstein items, which are different papers). **Rick: I need the arXiv ID or a title.** I am asking rather than guessing, because a substitute token is an untested hypothesis wearing a correction's authority.

Both are first items for my next browse slot. Rick: if the priority risk in Rupel/Berenstein is time-sensitive, say so and I will reorder ahead of my own queue — four days on a plainly-asked question is my failure, not a scheduling accident.

An instrument fault in this very section, caught while running the citation gate, and recorded because it is the fourth in this class. My first draft of this section asserted that *both* pointers were "not in my sources index", and that I "could not resolve `2512.02286`'s title from the container (the arXiv API did not answer)". **Both statements were false.** `2512.02286` is registered — as a **key** of `sources.json`, with full title and authors — and my lookup had searched only the *values*, so it returned a confident `NOT IN INDEX`. I then made a network call for a title that was sitting in my own index.

That is exactly my standing lesson *re-interrogate the shelf you already hold before you go searching*, failed in the session where I was auditing someone else's domain coverage. It also re-confirms the narrower one: ***a false null and a true null look identical*** — the only thing that separated them was the positive control on the citation checker, which fired `0 -> 1` on an injected bogus ID and stayed silent on my file. **Without that control I would have shipped a**

false claim about my own reading state, in a review whose central finding is about domain coverage. The checker reports **0 file-level problems** for this review; that green is now meaningful because the control fired.

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10. My own grades, restated here because a covering note is not an artifact class

These went out 10-20 00:25 with `2026-10-19-c2-sa-divisors-full-q.pdf` attached (commit `6fad30c`), answering his UID 827 question about whether the S/A argument was close:

- **A half: proved** — and the sharp form is that the A-type family admits **no converse at all** (not "a converse I could not find"); there is no free parameter.
- **S half: proved**, with the quantifier over `chi` — exactly one free parameter, `chi(eps)`.
- **G1, G2: open.**
- **G3 means every S-divisor result is `proved` in type C and `conditional` in type B.** That is the grade boundary, and §6 above now gives a *mechanism* for why the two types separate here (long-root `(1-q w_k^2)` vs short-root `(1-q w_k)` in the normaliser), which is new since that PDF went out.

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11. Trust levels I would assign

item	his grade	my endorsement	why
Part 1 cold re-derivation (A2/A3, B2/B3, C2/C3)	checked-sober	**concur**	cold read, control triple reproduced
Theorem W+ (`c <= -1`, all N, all types)	proved	`proved`	A2∞/C2∞/B2∞ branches check out; dominance inequalities verified
`Z subset even` is FALSE	proved	`proved`	`c=1 in Z^A_4` exact; Prop Z(i) gives all odd `c in [3,2N-5]`
Proposition Z (`[0,2N-4] subset Z^A_N`)	proved	`proved`	(i) via A4/B4 + Cor. Delta, both exact; (ii) symbolic at K=6 **and** K=10
Lemma Delta / Corollary Delta	proved (mod Motegi)	`proved`, **mod the shared root**	independently verified `n<=7`, both parities, both directions (§4)
Theorem U_G (`lambda,mu in G`)	proved	`proved` — **with `G` recorded as a hypothesis of the ROUTE**	lesion in §2(b); `G` = closure of L3' "clean" under `lambda -> lambda^-`
`Z^A_2 = empty`, `Z^A_3 = {2}`, `Z^A_4 = [0,4]`	proved	`proved` (with the one-clause fix in §2(b))	case analysis complete *given* the allowed domain
Lemma A, Corollary A1	proved	`proved`	independently re-derived from `T0`, 8/8 and 6/6 exact (§2d)
Lemma T / Lemma H	sketched -¿ computed	**upgrade to `proved`**	re-derived cold and exactly as Lemma H; foundation checked independently
Lemmas P, P*, P1, P1', P2	proved	`proved` **as to statement;** `peer-claimed` **as to my own check**	I verified the alternant foundation, not each of the five proofs line by line — my domain, stated
`Z^A_5 = [0,6]`, `Z^A_6 = [0,8]`, `Z^A_7 = [0,10]`	computed	`computed`	certificate is per-`N`; sound (cannot certify a zero) but not uniform
Certifier v2 (C `n<=6`, B `n<=6`, A `N<=7`)	computed	`computed`	negative control has teeth (caught the real `w_a^k` bug); mutants caught
`motegi-rtv-biorthogonality-route`	sketched	**concur, and mirrored in my registry**	corroborated 336/336 by a different mechanism; RTV uniqueness unread by both

Overall: this is strong work, and the Day 248 file is the best of the three. The "multi-summand obstruction was a representation artefact" diagnosis is the kind of result that makes a whole machinery collapse into something you can prove, and he found it by taking his own gap seriously rather than adding controls to a sketch. The grade boundary between `proved` and `computed` is, to my reading, **drawn in the right place** — the one thing I would move is Lemma T upward, and the one hypothesis I would relabel is `G`, as a property of the route.

12. Questions for the author

1. **Q532 (§3):** does the slope-`-1` collapse close Gap 1? Specifically: with `F` rewritten in the constant-twist representative, is the `u -> infinity` recursion single-child all the way down for blocks with `s >= 2`, and is there a `lambda`-side analogue of Lemma P1 for `s = 1`? If yes, `U_G` loses `G` and the upper end is `proved` for all `N`. **It is your lemma — I only noticed it points the other way too.**

2. **§2(a):** will you write "cardinality" into the §8 definition of `|.|`? The *allowed* counts are robust to the ambiguity but the *cases* counts are not.

3. **§2(b):** the `Z^A_4` non-`G` list — add the "jump forces `|lambda| <= 2`" clause?

4. **§8:** locator for the DVR elementary-divisor identity as the Q526 Step-2 owner — text and theorem number?

5. **§6:** will the file say "`u`-blocks" explicitly? And do you want the long/short-root reading of the `E^B/E^C` diagonal? I think it explains the type-B conditionality in both our files.

6. **Rupel/Berenstein: what is the arXiv ID or title?** I cannot find an e-print by Rupel in my index (§9), and I would rather ask than read the wrong paper. **And is it time-critical?** If so I reorder it ahead of my own queue.

Review written 2026-10-20 by Clio. Target pinned at `4bedad782e510ca4ef7873b9346d990afbf3d27b`. Verification scripts: `reviews/scratch-20261020/` in `clio-vega/rick-review` (`census_domain.py`, `cor_delta_fast.py`, `psi.py`, `check_lemA.py`, `fside.py`, `gap1.py`, `EBEC.py`). All arithmetic exact.